

Arithmetic Power Geometry Adds Interpretable Structural Information to Classical and Principal-Component Representations of Single-Cell PBMC Profiles

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Abstract

Single-cell analyses require representations that retain biological structure while remaining interpretable. Arithmetic Power Geometry (APG) describes a nonnegative molecular profile through normalized squared weights and an exponent-indexed deformation curve, but its value depends on showing complementarity rather than claiming to replace latent embeddings. We therefore rebuilt the evaluation around three tests: direct ablation against classical summaries, explicit comparison with principal-component representations, and validation using labels defined independently of APG. In the 10x Genomics Fresh 68k PBMC transcriptomic benchmark (68,579 cells), five-fold classification showed that programme-level classical summaries achieved macro-F1 0.2838 ± 0.0078 , programme APG achieved 0.3806 ± 0.0135 , and their combination reached 0.4127 ± 0.0108 . Across gene- and programme-level summaries, adding APG increased macro-F1 from 0.4258 ± 0.0100 to 0.5333 ± 0.0035 . High-exponent gene-level defects saturated, identifying $2 < p \lesssim 3$ as the useful transition regime. Independent validation used a compact Azimuth/Hao PBMC reference containing 36,433 cells, frozen eight-class annotations, 228 normalized antibody-derived features, and a 50-dimensional RNA reference PCA. With fold-specific 30-component ADT PCA, adding APG increased macro-F1 from 0.9382 ± 0.0017 to 0.9590 ± 0.0033 ; after classical ADT summaries were included, APG still increased macro-F1 from 0.9468 ± 0.0026 to 0.9594 ± 0.0030 . These results support APG as a compact, interpretable structural layer that contributes measurable information beyond matched classical summaries and augments conventional principal-component representations without claiming to reconstruct the cell-state manifold.

Keywords: Arithmetic Power Geometry; single-cell analysis; PBMC; interpretable descriptors; exponent deformation; multimodal validation; PCA

1 Introduction

Single-cell RNA sequencing (scRNA-seq) resolves molecular heterogeneity at cellular scale, but its analytical value depends on representations that reduce sparsity and dimensionality without erasing biologically relevant structure. Contemporary workflows rely on principal-component analysis (PCA), graph-based neighbourhood representations, nonlinear visualizations, and probabilistic latent-variable models such as scVI (Heumos et al., 2023; Lopez et al., 2018; Luecken & Theis, 2019; Stuart et al., 2019; Wolf et al., 2018). These methods are powerful precisely because they preserve multivariate relationships needed for clustering, integration, annotation, and visualization. Their coordinates, however, are not generally reducible to a small set of transparent distributional quantities.

A complementary line of analysis summarizes each cell by diversity, concentration, or inequality statistics. Shannon entropy, Rényi entropy, Simpson diversity, Herfindahl concentration, Gini inequality, and participation ratios are interpretable because each has an explicit mathematical meaning (Gini, 1912; Lloyd & Ghelardi, 1964; Rényi, 1961; Shannon, 1948; Simpson, 1949). Their limitation is different from that of latent embeddings: a single scalar reports only one aspect of how molecular mass is distributed across coordinates. A useful compact descriptor should therefore not be evaluated as a replacement for a cell-state manifold. The more relevant question is whether it contributes reproducible structural information beyond classical summaries and whether that information remains useful when attached to an established latent representation.

Arithmetic Power Geometry (APG) was developed around this narrower objective. APG first maps a nonnegative feature vector to normalized squared weights and then studies how those weights respond to an exponent deformation (Akhtar, 2026a, 2026b). The resulting defect $D(p)$ is bounded, monotone for $p \geq 2$, and locally controlled by entropy at the Euclidean point. Thus APG is explicitly a compressed representation, but one that exposes a response trajectory rather than only a static statistic. That distinction is mathematically clear; its empirical usefulness must nevertheless be demonstrated by ablation, independent labels, and direct comparison with latent features.

Biologically meaningful aggregation provides a natural intermediate scale for such tests. Gene-set and regulatory-programme methods such as SCENIC and UCell quantify coordinated activity at single-cell resolution (Aibar et al., 2017; Andreatta & Carmona, 2021). Recent benchmarking has shown that pathway scores depend on gene-set composition, sparsity, noise, and scoring design (Wang & Thakar, 2024); updated implementations continue to emphasize robust rank-based scoring and reproducibility (Andreatta & Carmona, 2026). More broadly, recent multimodal benchmarks show that no representation is uniformly optimal across dimensionality reduction, classification, clustering, integration, and robustness tasks (Fu et al., 2025; Liu et al., 2025). These observations motivate testing APG as an augmenting structural layer rather than as a universal embedding.

The present study therefore addresses five concrete questions. First, do gene-level APG descriptors add discrimination beyond classical cell-level summaries in a large sparse PBMC transcriptomic benchmark? Second, after genes have already been aggregated into biological programmes, does APG add information beyond classical summaries computed from the *same* programme profile? Third, which portion of the exponent curve remains informative before high-dimensional saturation? Fourth, does APG augment a conventional PCA representation on an independent reference object whose labels were defined without reference to APG? Fifth, are the principal findings robust to count thinning and to technical partitioning? The two data resources and their distinct roles are summarized in table 1.

2 Materials and methods

2.1 Study design and data resources

The primary transcriptomic benchmark was the official 10x Genomics Fresh 68k PBMC dataset from Donor A (Zheng et al., 2017). The filtered matrix contained 68,579 cells and 32,738 genes. The Cell Ranger summary reported 20,491 mean reads per cell, 525 median detected genes, and 1,292 median UMI counts. Raw UMI counts were divided by each cell’s library total, multiplied by 10,000, and transformed as $z_{ci} = \log(1 + x_{ci})$.

External validation used the supplied Azimuth Human PBMC reference object (Satija Lab, 2021), which derives from the multimodal PBMC atlas of Hao et al. (2021). The full Azimuth PBMC reference is based on 24 samples from eight volunteers and jointly models RNA and

antibody-derived tag (ADT) information. The compact reference object analysed here contained 36,433 cells, 228 nonnegative normalized ADT features, a 50-dimensional RNA reference PCA (the `refDR` embedding), and frozen hierarchical annotations. We used the broad `celltype.11` labels: CD4 T (17,646), CD8 T (5,858), monocyte (4,564), B cell (2,698), NK (2,051), other T (1,616), other (1,078), and dendritic cell (922). These annotations were not generated from APG and were not altered during the analysis. The compact object’s RNA count/data matrix is intentionally absent, so external APG was computed on the 228-dimensional nonnegative ADT profile; the supplied 50-dimensional RNA PCA was used as an orthogonal latent baseline. This distinction prevents reconstruction or fabrication of transcriptomic expression that is not present in the reference object.

Table 1: Data resources and analytical roles. The Azimuth object used here is a compact reference object; it is not the full 161,764-cell reference described by the public Azimuth portal.

Resource	Cells/features	Labels	Role in this study
10x Fresh 68k PBMC	68,579 cells; 32,738 genes	Eight provisional marker-supported classes plus uncertain cells	Primary transcriptomic benchmark; gene- and programme-level APG; ablation, saturation, clustering, and count-thinning analyses
Azimuth/Hao compact PBMC reference	36,433 cells; 228 ADT features; 50 RNA PCs	Frozen <code>celltype.11</code> eight-class annotations	Independent-label validation; fold-specific ADT PCA versus PCA+APG; RNA-PCA augmentation; technical group sensitivity

2.2 Classical distributional descriptors

For cell c , let $z_{ci} \geq 0$ denote a normalized molecular feature value over m coordinates. Ordinary weights are

$$q_{ci} = \frac{z_{ci}}{\sum_{j=1}^m z_{cj}}, \quad \sum_i q_{ci} = 1. \quad (1)$$

Classical descriptors were Shannon entropy $-\sum_i q_{ci} \log q_{ci}$, Rényi-2 entropy $-\log \sum_i q_{ci}^2$, Simpson diversity $1/\sum_i q_{ci}^2$, Herfindahl concentration $\sum_i q_{ci}^2$, participation ratio $(\sum_i q_{ci}^2)^{-1}$, Gini inequality, and detected-feature count. The same definitions were used for gene-level expression, programme profiles, and external ADT profiles wherever applicable.

2.3 APG weights and deformation profile

APG uses normalized squared weights

$$w_{ci} = \frac{z_{ci}^2}{\sum_{j=1}^m z_{cj}^2}, \quad \sum_i w_{ci} = 1, \quad (2)$$

with entropy and normalized entropy

$$H_{\text{APG}}(c) = -\sum_i w_{ci} \log w_{ci}, \quad \tilde{H}_{\text{APG}}(c) = \frac{H_{\text{APG}}(c)}{\log m}. \quad (3)$$

For $p \geq 2$, the APG defect is

$$D_c(p) = 1 - \sum_i w_{ci}^{p/2}. \quad (4)$$

It follows immediately that $D_c(2) = 0$ and $0 \leq D_c(p) < 1$ for $p > 2$. Around the Euclidean point,

$$D_c(2 + \varepsilon) = \frac{\varepsilon}{2} H_{\text{APG}}(c) + O(\varepsilon^2), \quad (5)$$

so entropy fixes the first-order deformation rate. We evaluated $p \in \{2, 2.25, 2.5, 3, 4, 5, 6\}$ and integrated deformation

$$\mathcal{A}_c(2, 6) = \int_2^6 D_c(p) dp, \quad (6)$$

using trapezoidal quadrature. Maximum concentration, quadratic concentration, effective dimension $\exp(H_{\text{APG}})$, and APG participation ratio were also retained.

2.4 Provisional transcriptomic labels and biological programmes

For the Fresh 68k benchmark, eight deterministic marker modules represented CD4 T, CD8 T, NK, B cell, CD14 monocyte, FCGR3A monocyte, dendritic, and platelet populations. A cell was assigned only when its highest standardized module score was at least 0.25 and exceeded the second-highest score by at least 0.10; otherwise it was retained as uncertain. This procedure yielded 56,903 confident cells (83.0%) and 11,676 uncertain cells (17.0%). The labels are explicitly treated as internal computational annotations because marker choice and thresholding can affect cell identification (Abdelaal et al., 2019; Pullin et al., 2024).

Nine programme scores were computed as mean normalized expression over predefined sets representing T-cell receptor signalling, cytotoxicity, B-cell receptor activity, interferon response, inflammatory myeloid activity, oxidative phosphorylation, ribosomal activity, cell cycle, and apoptosis/stress (table 5). Programme APG applies equations (2) to (4) and (6) to the resulting nine-dimensional nonnegative programme vector. Importantly, the strengthened ablation compares APG with classical diversity/concentration summaries computed from this *same programme vector*, thereby separating the effect of programme aggregation from the additional effect of APG transformation.

2.5 Fresh 68k ablation and statistical analyses

A balanced subset of 6,400 confidently annotated cells (800 per provisional class) was sampled reproducibly. Seven representations were compared under identical five-fold stratified splits: gene classical; gene classical+APG; programme classical; programme APG; programme classical+APG; all classical (gene+programme classical); and all classical+all APG. Features were standardized using training-fold means and standard deviations. Classification used class-balanced one-versus-rest logistic regression, with macro-F1 and balanced accuracy as primary metrics. This ablation is not interpreted as external annotation accuracy because the provisional labels and biological programmes share immune-marker information.

Between-class effects were assessed by Kruskal–Wallis tests (Kruskal & Wallis, 1952) and epsilon-squared

$$\epsilon^2 = \max \left\{ 0, \frac{H - k + 1}{N - k} \right\}, \quad (7)$$

where H is the Kruskal–Wallis statistic, k the number of groups, and N the number of cells. Because large cell counts make extremely small p values routine, effect size rather than significance was emphasized.

To quantify how much the exponent profile departs from the entropy-controlled local regime, each $D(p)$ was regressed on normalized APG entropy across confident cells and the resulting R^2 was reported. This is a deliberately stringent redundancy diagnostic: high R^2 near $p = 2$ is expected from equation (5); decreasing R^2 at larger p indicates departure from the first-order entropy relationship, while simultaneously shrinking dynamic range can signal saturation.

2.6 External Azimuth/Hao validation against PCA

The external reference analysis used the frozen `celltype.11` labels and the 228-dimensional nonnegative ADT matrix stored in the supplied reference object. Classical and APG descriptors were computed directly from each ADT profile. For each of five stratified folds, the 228 ADT variables were standardized using training data only, and a 30-component PCA was fitted on the training partition and applied to the held-out partition. This fold-specific construction prevents label leakage through preprocessing.

A class-balanced ridge classifier was then evaluated on: ADT PCA; ADT PCA+classical descriptors; ADT PCA+APG; and ADT PCA+classical+APG. In parallel, the published 50-dimensional RNA `refDR` representation was tested alone and after augmentation with ADT classical descriptors, ADT APG, or both. The RNA PCA embedding is a fixed, unsupervised reference feature already contained in the Azimuth object; it was not refitted from labels. All classifier inputs were standardized within training folds. Because ADT information is an additional modality relative to RNA PCA, comparisons involving RNA PCA+ADT features are interpreted as cross-modal complementarity; the same-modality ADT PCA ablation is the cleaner test of whether APG adds information beyond a PCA representation of the same molecular profile.

Cell identifiers contained 13 reproducible barcode prefixes. As a secondary technical sensitivity analysis, each prefix group was held out in turn and ridge classification was repeated. These prefixes are *not* interpreted as donor identifiers. They provide a grouped technical perturbation only; no donor-level biological inference is made from them.

2.7 Count-depth robustness and cluster-separation boundary

For 3,000 Fresh 68k cells, raw UMI counts were thinned binomially to 80%, 60%, 40%, and 20% of original depth. The complete normalization and descriptor pipeline was rerun, and Spearman correlation with each cell’s full-depth value quantified rank stability. Separately, Euclidean silhouette coefficients (Rousseeuw, 1987) were evaluated on standardized compact descriptor spaces. Negative silhouettes are interpreted as evidence that APG summaries should not be treated as a replacement cell-state manifold.

2.8 Software and reproducibility

The original Fresh 68k pipeline and derived cell-level descriptor table were supplied as part of the reproducibility archive. New ablation and external-reference analyses were executed in Python using NumPy, pandas, SciPy, and scikit-learn (Pedregosa et al., 2011). The final package contains the analysis scripts, exact result tables, source figures, manuscript, and checksums information. The Azimuth reference is cited as a data resource (Satija Lab, 2021); the broader single-cell computational ecosystem is described by Virshup et al. (2023).

3 Results

3.1 Programme-level APG showed the largest class-associated effects

The original Fresh 68k effect-size analysis showed that programme-level APG descriptors had substantially larger class-associated effects than gene-level compact summaries. Programme APG maximum concentration yielded $\epsilon^2 = 0.563$, integrated deformation 0.541, $D(3)$ 0.526, and normalized APG entropy 0.493. The largest gene-level APG effect was $D(6)$ at 0.169, whereas

Shannon entropy, Rényi-2 entropy, Simpson diversity, Herfindahl concentration, and ordinary participation ratio were each approximately 0.081. These results indicate that biologically named programme aggregation exposes strong class structure, but they do not by themselves show that APG—rather than programme construction—causes the gain. That question is addressed directly by the ablation in section 3.3.

3.2 High exponents depart from entropy but simultaneously saturate

Gene-level deformation profiles rose sharply between $p = 2$ and $p = 3$ and were nearly flat thereafter (figure 1). Consistent with equation (5), normalized APG entropy explained 99.85% of the linear variance of $D(2.25)$ and 99.42% of $D(2.5)$. The relationship weakened progressively at larger exponents: $R^2 = 0.9784$ for $D(3)$, 0.9277 for $D(4)$, 0.8676 for $D(5)$, and 0.8089 for $D(6)$ (figure 2). However, the standard deviation of $D(p)$ simultaneously collapsed from 0.01365 at $p = 2.25$ to 4.08×10^{-6} at $p = 6$. Thus larger exponents are mathematically less reducible to entropy but empirically less useful because their dynamic range is almost exhausted. The informative compromise lies in the transition region around $2 < p \lesssim 3$, rather than at the highest tested powers.

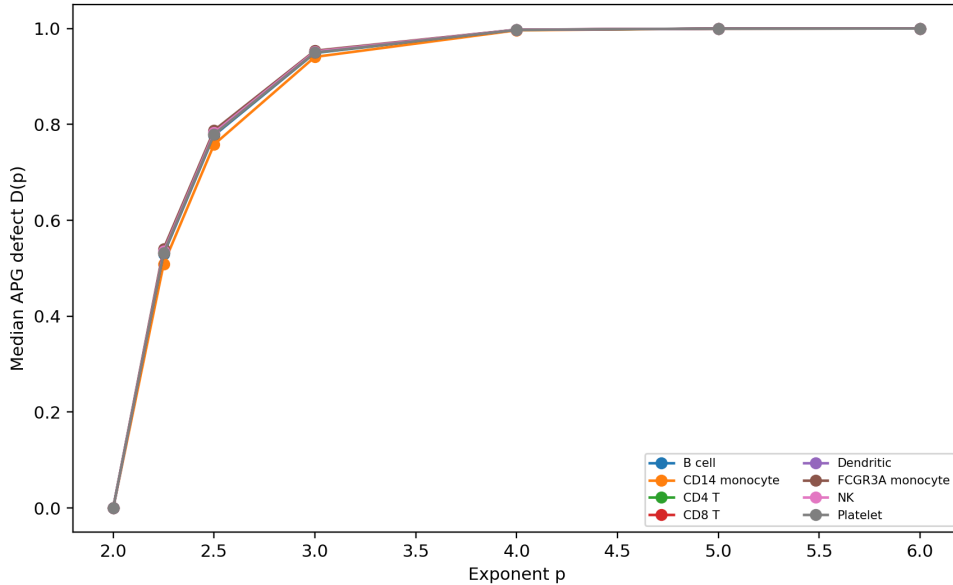


Figure 1: Median gene-level APG deformation profiles across provisional Fresh 68k PBMC classes. The curves rise most rapidly between $p = 2$ and $p = 3$ and approach saturation thereafter.

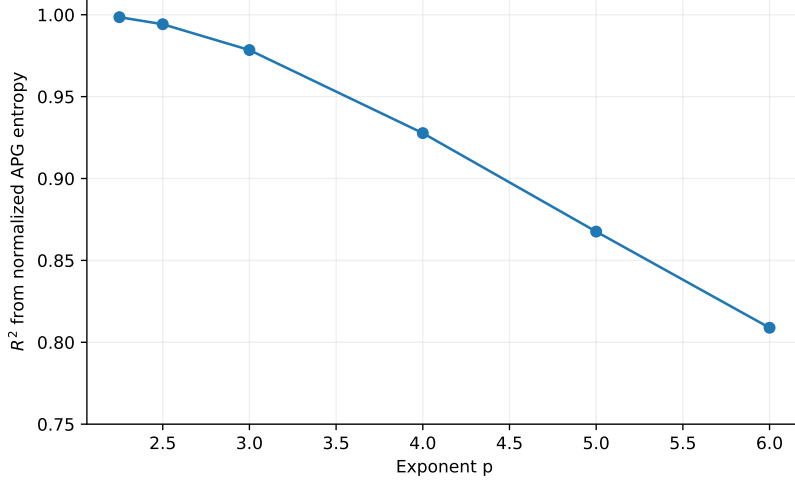


Figure 2: Linear redundancy of gene-level APG defects with normalized APG entropy. The local regime is strongly entropy-controlled, as predicted by equation (5); R^2 decreases with exponent, but high- p defects simultaneously lose dynamic range through saturation.

3.3 Programme ablation separates APG from programme aggregation

The dedicated ablation directly compared classical and APG summaries of the same nine-dimensional programme vector (table 2 and figure 3). Programme classical descriptors achieved macro-F1 0.2838 ± 0.0078 ; programme APG alone achieved 0.3806 ± 0.0135 ; and programme classical+APG reached 0.4127 ± 0.0108 . Thus APG increased macro-F1 by 0.1289 over classical summaries of the same programme aggregation. At gene level, adding APG increased macro-F1 from 0.3274 ± 0.0059 to 0.3876 ± 0.0041 . When gene- and programme-level classical summaries were combined, macro-F1 was 0.4258 ± 0.0100 ; adding all APG descriptors increased it to 0.5333 ± 0.0035 , an absolute gain of 0.1074. All five paired folds favored the APG-augmented representation in each of these three contrasts; with only five folds, the exact two-sided Wilcoxon value was $p = 0.0625$ and is not treated as confirmatory inference.

Table 2: Fresh 68k five-fold ablation on a balanced 6,400-cell subset. Values are mean $\pm$ standard deviation across identical folds. Programme classical and programme APG are computed from the same nine programme scores, isolating the effect of APG transformation from programme aggregation.

Representation	Features	Balanced accuracy	Macro-F1
Programme classical	7	0.3330 ± 0.0085	0.2838 ± 0.0078
Programme APG	12	0.4178 ± 0.0119	0.3806 ± 0.0135
Programme classical + APG	19	0.4303 ± 0.0080	0.4127 ± 0.0108
Gene classical	7	0.3722 ± 0.0070	0.3274 ± 0.0059
Gene classical + APG	19	0.4131 ± 0.0038	0.3876 ± 0.0041
All classical	14	0.4538 ± 0.0110	0.4258 ± 0.0100
All classical + all APG	38	0.5450 ± 0.0026	0.5333 ± 0.0035

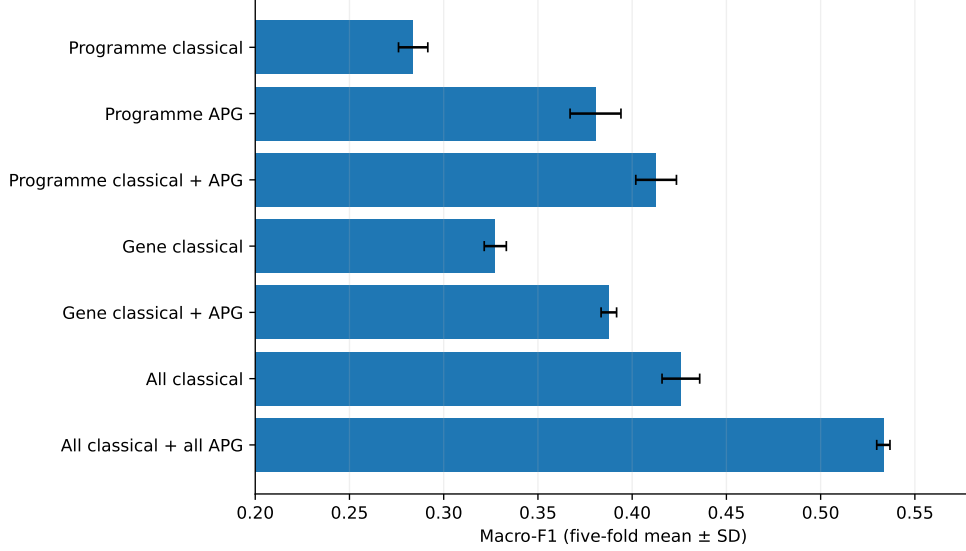


Figure 3: Fresh 68k ablation. Programme-level classical summaries, programme APG, and their combination are computed from the same biological programme profile. Error bars show five-fold standard deviations.

3.4 Independent Azimuth labels support APG augmentation of PCA

The independent reference analysis directly addressed the question of whether APG can add information to a conventional latent representation. On the same 228-dimensional ADT molecular profile, fold-specific 30-component PCA achieved macro-F1 0.9382 ± 0.0017 . Adding APG increased macro-F1 to 0.9590 ± 0.0033 , a mean gain of 0.0208 across the five held-out folds (table 3 and figure 4). Classical ADT summaries alone also augmented PCA, reaching 0.9468 ± 0.0026 ; adding APG on top of those classical summaries raised macro-F1 further to 0.9594 ± 0.0030 , a gain of 0.0125. The corresponding balanced-accuracy values increased from 0.9359 ± 0.0034 for ADT PCA to 0.9624 ± 0.0028 for ADT PCA+APG and 0.9625 ± 0.0023 for ADT PCA+classical+APG.

The fixed 50-dimensional RNA reference PCA produced macro-F1 0.9429 ± 0.0035 . Adding ADT APG yielded 0.9539 ± 0.0038 ; adding classical ADT summaries yielded 0.9475 ± 0.0034 ; and adding both reached 0.9548 ± 0.0036 . Because these RNA-PCA augmentations introduce an additional modality, they demonstrate cross-modal complementarity rather than an APG-only advantage. The same-modality ADT PCA comparisons are therefore the primary external test.

Table 3: External Azimuth/Hao validation with frozen broad cell-type labels. ADT PCA was fitted within each training fold; values are five-fold mean $\pm$ standard deviation.

Representation	Features	Balanced accuracy	Macro-F1
ADT PCA	30	0.9359 ± 0.0034	0.9382 ± 0.0017
ADT PCA + classical	37	0.9458 ± 0.0016	0.9468 ± 0.0026
ADT PCA + APG	42	0.9624 ± 0.0028	0.9590 ± 0.0033
ADT PCA + classical + APG	49	0.9625 ± 0.0023	0.9594 ± 0.0030
RNA PCA	50	0.9465 ± 0.0038	0.9429 ± 0.0035
RNA PCA + ADT classical	57	0.9508 ± 0.0041	0.9475 ± 0.0034
RNA PCA + ADT APG	62	0.9568 ± 0.0045	0.9539 ± 0.0038
RNA PCA + ADT classical + APG	69	0.9579 ± 0.0042	0.9548 ± 0.0036

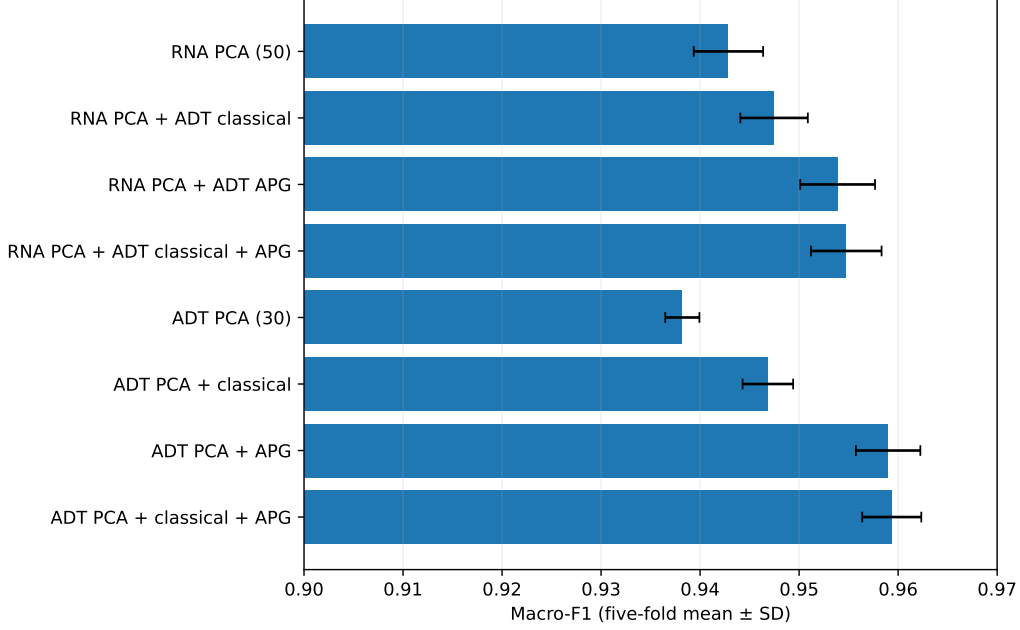


Figure 4: Independent-label Azimuth/Hao validation. The principal same-modality comparison is ADT PCA versus ADT PCA augmented with classical and/or APG descriptors. RNA PCA combinations provide a separate cross-modal complementarity analysis.

3.5 Grouped technical sensitivity preserved the augmentation pattern

When each of the 13 barcode-prefix groups was held out in turn, the same ordering persisted (table 4). ADT PCA achieved mean macro-F1 0.9336 ± 0.0135 across groups, while ADT PCA+APG achieved 0.9569 ± 0.0074 (mean paired gain 0.0233). After classical summaries were included, APG increased group-held-out macro-F1 from 0.9427 ± 0.0109 to 0.9571 ± 0.0072 (gain 0.0144). All 13 paired group differences favored APG in these comparisons; the exact paired Wilcoxon value was $p = 0.000244$. These groups are technical barcode prefixes, not verified donors, so this result is a robustness check rather than biological replication.

Table 4: Technical barcode-prefix held-out sensitivity on the compact Azimuth reference. The 13 groups are not interpreted as donors.

Representation	Balanced accuracy	Macro-F1
ADT PCA	0.9311 ± 0.0153	0.9336 ± 0.0135
ADT PCA + classical	0.9414 ± 0.0133	0.9427 ± 0.0109
ADT PCA + APG	0.9594 ± 0.0088	0.9569 ± 0.0074
ADT PCA + classical + APG	0.9592 ± 0.0084	0.9571 ± 0.0072
RNA PCA	0.9467 ± 0.0059	0.9422 ± 0.0057
RNA PCA + ADT APG	0.9559 ± 0.0049	0.9527 ± 0.0053

3.6 Count thinning favors low-to-moderate exponents

The Fresh 68k depth perturbation independently supported the low-exponent interpretation. At 20% retained counts, Shannon entropy retained Spearman $\rho = 0.951$, normalized APG entropy 0.917, $D(3)$ 0.878, and integrated deformation 0.893, whereas $D(6)$ fell to 0.770 (figure 5). At 40% retention, the corresponding APG correlations were 0.966, 0.946, and 0.954 for normalized entropy, $D(3)$, and integrated deformation. APG was therefore robust to substantial thinning at

low-to-moderate exponents, but it was not uniformly more stable than Shannon entropy; high powers were clearly more sensitive to count perturbation.

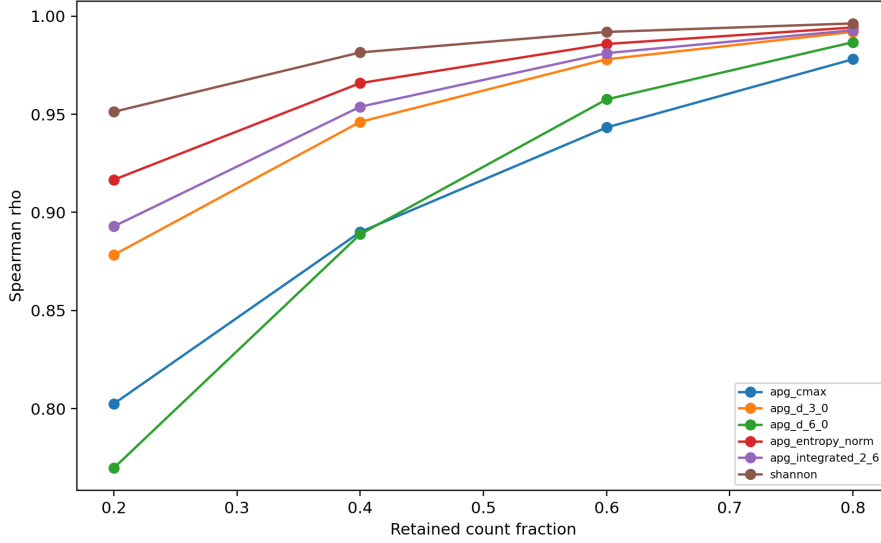


Figure 5: Fresh 68k rank stability under binomial thinning of raw UMI counts. Curves show Spearman correlation with each descriptor’s full-depth value. High-exponent $D(6)$ is less stable than lower-exponent and entropy-based descriptors.

3.7 Entropy-matched cells can retain distinct deformation structure

A targeted case analysis illustrates the distinction between a static diversity statistic and the APG trajectory without treating it as independent biological validation. Within the provisional CD4 T population, two cells had nearly identical ordinary Shannon entropy (6.6980 and 6.6974; absolute difference 0.00066) but $D(3)$ values of 0.9604 and 0.9538 and integrated deformation values of 3.6594 and 3.6455. Their normalized APG entropies also differed (0.6315 versus 0.6128), demonstrating that squaring the expression weights changes the structural emphasis even when ordinary Shannon diversity is essentially matched. Their marker-module patterns were not identical: the first cell had CD4 and B-cell scores of 1.379 and 0.636 with no CD8/NK signal, whereas the second had a stronger CD4 score (1.810) together with detectable CD8 (0.910), NK (0.387), B-cell (0.431), and dendritic (0.234) module signals. This example does not establish a distinct biological subtype; it demonstrates concretely that near-equal ordinary Shannon diversity can coexist with different squared-weight concentration/deformation structure and different marker-supported composition.

3.8 Negative silhouettes define a boundary rather than a contradiction

All compact descriptor spaces in the Fresh 68k analysis had negative Euclidean silhouette coefficients: the combined classical+gene+programme APG space was least negative at -0.096 , followed by classical+gene APG at -0.143 , while programme APG alone was -0.214 . These results show that compact APG descriptors do not reconstruct the neighbourhood geometry required of a general cell-state embedding. In light of the external PCA augmentation results, this failure becomes informative: APG can improve a downstream linear classification representation while remaining unsuitable as a stand-alone manifold. This is consistent with recent benchmark evidence that different single-cell representations should be judged on task-specific objectives rather than on a single universal criterion (Fu et al., 2025; Liu et al., 2025).

4 Discussion

This study changes the empirical status of APG in three ways. First, it replaces a broad embedding-replacement narrative with a falsifiable complementarity claim. Second, it separates biological programme aggregation from the APG transform through an explicit same-input ablation. Third, it introduces an independent reference benchmark in which APG is tested alongside PCA rather than only against classical scalar descriptors.

The programme ablation is particularly important. Strong programme-level class effects could otherwise be attributed entirely to the act of collapsing genes into biologically named modules. That explanation is only partly sufficient. Classical diversity/concentration summaries of the same nine-dimensional programme vector produced macro-F1 0.2838, whereas APG descriptors of that vector produced 0.3806 and their combination produced 0.4127. APG therefore contributes a measurable transformation-specific signal beyond aggregation itself. The result remains an internal benchmark because marker-derived targets and programmes share biological vocabulary; accordingly, the stronger evidence for general complementarity comes from the independent Azimuth labels.

The external reference results directly connect APG to conventional latent representations. In a same-modality test, fold-specific ADT PCA improved from macro-F1 0.9382 to 0.9590 when APG descriptors were appended. The improvement remained after seven classical ADT summaries were already present, rising from 0.9468 to 0.9594. This is the most direct evidence in the study that APG is not merely duplicating a standard compact statistic or substituting one compression for another. The PCA retains multivariate directions; APG adds explicit information about concentration and exponent response. Their combination performs better than either structural summary alone under the tested linear classifier.

The RNA-PCA analysis provides a second, intentionally different perspective. Adding ADT APG to the supplied RNA PCA improves classification, but this gain includes genuine cross-modal information and therefore cannot be attributed solely to the APG transform. The comparison with RNA PCA+ADT classical summaries partially controls this issue: adding APG on top of RNA PCA+classical ADT summaries still increased macro-F1 by approximately 0.0073. Nevertheless, the same-modality ADT PCA experiment remains the cleaner mechanistic comparison. A future extension should repeat the identical PCA-versus-PCA+APG experiment on an external object that retains the full nonnegative RNA expression matrix.

The exponent analysis also resolves an apparent tension in APG. Near $p = 2$, the defect is almost completely determined by entropy, exactly as the local expansion predicts. Moving away from $p = 2$ reduces this redundancy, but large exponents force the defect toward one in high-dimensional profiles. Thus novelty and numerical usefulness pull in opposite directions. The useful regime is not “higher p is better”; it is the transition window in which the curve has begun to depart from first-order entropy control but has not yet saturated. In the present data, that window is concentrated roughly between $p = 2$ and $p = 3$. This observation argues for dense low-exponent sampling in future APG applications and against relying on $D(6)$ as a primary gene-level descriptor.

The entropy-matched CD4 T example provides a limited but concrete biological interpretation. Two cells can have almost indistinguishable Shannon diversity while differing in APG deformation and in the balance of marker-module signals. The example should not be exaggerated: it is selected from an internally annotated donor and does not establish a new CD4 T subtype. Its role is conceptual—to show what information the deformation profile can express that an ordinary entropy scalar cannot. Stronger biological validation should use independently annotated perturbation or disease cohorts and test whether APG changes correspond to reproducible pathway or state transitions.

Several limitations remain. The Fresh 68k transcriptomic benchmark contains one donor and uses provisional marker-supported labels; cell-level cross-validation is therefore not biological replication. The compact Azimuth object does not retain the RNA expression matrix, so external APG was necessarily evaluated on its 228-dimensional ADT profile rather than on transcriptomic counts. Although the broad Azimuth annotations are frozen independently of APG, they were originally constructed from the multimodal reference and are not equivalent to blinded expert adjudication. The 13 barcode-prefix groups used for sensitivity testing are technical groups, not donor identifiers. The external classifier is deliberately linear; performance under nonlinear learners may differ. Finally, many APG coordinates are mathematically related, and low-exponent defects are highly correlated with APG entropy. The contribution is therefore not feature independence in a strict statistical sense, but a reproducible combination of interpretable structural summaries whose incremental value can be directly measured.

These limitations also define clear next experiments. A fully retained multi-donor RNA matrix with frozen annotations would permit donor-held-out transcriptomic PCA versus PCA+APG evaluation, mixed-effects or pseudobulk inference where condition-level claims are made, and comparison with scVI or other latent models. Programme-level work should replace small manually curated modules with frozen pathway resources and benchmark APG augmentation of established scoring methods such as UCell (Andreatta & Carmona, 2021, 2026) while respecting known scoring sensitivities (Wang & Thakar, 2024). The present analysis provides the ablation logic and independent-reference design needed for those tests.

5 Conclusion

APG is best understood as an interpretable structural layer, not as a replacement for PCA, UMAP, scVI, or other cell-state representations. On the Fresh 68k transcriptomic benchmark, APG added discrimination beyond classical gene-level summaries and, critically, beyond classical summaries of the same biological programme aggregation. The exponent analysis showed that high-dimensional gene profiles saturate rapidly, concentrating useful deformation information near the low-to-moderate exponent transition. On an independent compact Azimuth/Hao PBMC reference, APG computed from nonnegative ADT profiles improved a fold-specific PCA representation and retained incremental value after classical summaries were included. Together, these findings support a narrower but stronger claim: exponent-deformation descriptors provide reproducible, interpretable structural information that can complement conventional single-cell representations when their scope and saturation limits are made explicit.

Data and code availability

The Fresh 68k PBMC dataset is publicly available from 10x Genomics (Zheng et al., 2017), and the Human PBMC reference is available through Azimuth/Zenodo (DOI: [10.5281/zenodo.4546839](https://doi.org/10.5281/zenodo.4546839)) (Hao et al., 2021). All code, derived data, statistical analyses, tables, and figures required to reproduce the study are available at [GitHub](#) and archived on Zenodo (DOI: [10.5281/zenodo.21306239](https://doi.org/10.5281/zenodo.21306239)).

Ethics statement

This study is a secondary computational analysis of publicly released, de-identified benchmark data. It involved no new participant recruitment, intervention, or collection of identifiable human information.

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Competing interests

The author declares no competing interests.

Author contributions

Mohammad Amir Khusru Akhtar: Conceptualization, methodology, software, formal analysis, validation, visualization, data curation, writing—original draft, writing—review and editing.

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A Supplementary programme definitions

Table 5: Curated programmes used in the Fresh 68k programme-level benchmark. This table is cited in the programme construction described in the Methods.

Programme	Genes
T-cell receptor signalling	CD3D, CD3E, CD3G, TRAC, LCK, LAT, ZAP70, CD247
Cytotoxicity	NKG7, GNLY, PRF1, GZMB, GZMH, CTSW, CCL5
B-cell receptor activity	MS4A1, CD79A, CD79B, CD37, CD74, HLA-DRA
Interferon response	ISG15, IFIT1, IFIT2, IFIT3, IFI6, MX1, OAS1, STAT1
Inflammatory myeloid	LYZ, S100A8, S100A9, FCN1, CTSS, TYROBP, LST1, LGALS3
Oxidative phosphorylation	NDUFA1, NDUFB1, NDUFS5, COX5A, COX6A1, ATP5F1, ATP5G1, UQCRB
Ribosomal activity	RPL3, RPL13, RPL32, RPS3A, RPS6, RPS12, RPS18, RPLP0
Cell cycle	MKI67, PCNA, STMN1, TUBA1B, TYMS, MCM2, MCM5, TOP2A
Apoptosis/stress	BAX, BCL2, FOS, JUN, DUSP1, HSPA1A, HSP90AA1, IER2

The programme definitions in table 5 were fixed before the strengthened ablation. They are transparent compact modules rather than claims of pathway-enrichment equivalence to SCENIC or UCell (Aibar et al., 2017; Andreatta & Carmona, 2021, 2026).